

Unit 117 - Field Data Review

Mohawk Valley West Slope demonstration. Method **Tractor**, 82.2 ac, BAF 20 variable-radius cruise on a 300 ft grid, 44 plots, 490 trees tallied. Simulated data with planted recording errors; see docs/methods.md.

Table 1. Stand summary from the BAF 20 cruise, per acre

STAND DENSITY		STAND INFO	
Basal area	223 sq ft/ac (SE 2.0 %)	Type	Sierran mixed conifer
Trees per acre	89	CWHR size / density	4 / D (cover 88 %)
QMD	21.4 in	SDI	286 (49 % of max 580)
SAWTIMBER (>= 10 in)		BIOMASS (< 10 in)	
BA QMD	219 sq ft/ac 22.4 in	BA QMD	4 sq ft/ac 8.7 in
TPA	80	TPA	9
Volume	46.8 MBF/ac (8507 cu ft)	Volume	2.0 green tons/ac (67 cu ft)
TREATMENT TARGET		CRUISE DESIGN	
Leave	35 % of max SDI = ~158 sq ft/ac BA	Sampling error	4.0 % at 95 % (t = 2.02, n = 44)
Species (BA)	WF 69, PP 66, IC 36, DF 33, SP 18	Stratum standard	40 % - MEETS; 44 plots vs 20-plot practice: ok

Cubic volume by form factor 0.42 net of defect; Scribner board feet at 5.5 bf/cu ft (Keegan et al. 2010); green tons at 60 lb/cu ft. SDI in the summation form; maximum is the basal-area-weighted FVS Western Sierra species maximum. Sampling error at 95 % confidence with Student's t; stratum standard 40 % for tree-measurement sales (FSH 2409.12 ch. 40, 41.1).

Table 2. Stand table by 4-inch DBH class, per acre

DBH class, in	Trees	TPA	BA	cu ft	MBF
4-7.9	1	1.4	0.5	7	0.0
8-11.9	20	16.0	9.1	194	1.1
12-15.9	40	17.3	18.2	482	2.6
16-19.9	67	17.1	30.5	920	5.1
20-23.9	83	14.2	37.7	1325	7.3
24-27.9	71	8.9	32.3	1247	6.9
28-31.9	66	6.2	30.0	1264	7.0
32-35.9	50	3.6	22.7	1007	5.5
36-39.9	31	1.8	14.1	657	3.6
40-43.9	21	1.0	9.5	483	2.7
44-47.9	19	0.7	8.6	450	2.5
48-51.9	9	0.3	4.1	225	1.2
52-55.9	7	0.2	3.2	176	1.0
56-59.9	3	0.1	1.4	85	0.5
60-63.9	2	0.0	0.9	52	0.3
All	490	89.0	222.7	8574	47.2

Table 3. Stock table by species, per acre

Species	Trees	TPA	BA	QMD	cu ft	MBF
WF	151	26.6	68.6	21.8	2674	14.7
PP	146	28.4	66.4	20.7	2531	13.9
IC	80	15.3	36.4	20.8	1412	7.8
DF	73	12.5	33.2	22.0	1251	6.9
SP	40	6.1	18.2	23.3	706	3.9
All	490	89.0	222.7	21.4	8574	47.2

Trees = trees tallied in the cruise; TPA, BA (sq ft/ac), cubic feet and Scribner MBF per acre by BAF expansion. Species: PP ponderosa pine, WF white fir, DF Douglas-fir, SP sugar pine, IC incense-cedar.

Unit 117: plots on the 300 ft grid (orange = QA flag)

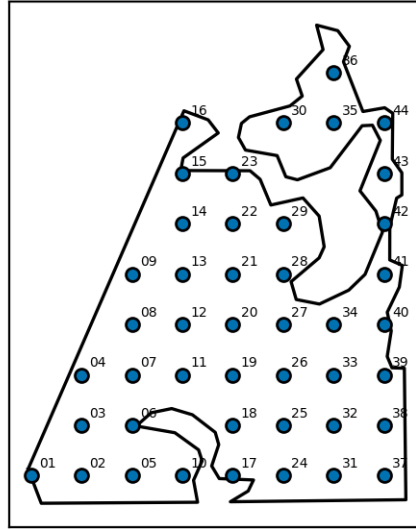


Figure 1. Plot locations with QA status.

QA findings: no issues found; unit accepted

Table 4. Plot list

Plot	Trees in	BA sq ft/ac	Slope %	Status
117-01	11	220	17	ok
117-02	12	240	201	ok
117-03	13	260	85	ok
117-04	13	260	35	ok
117-05	10	200	94	ok
117-06	10	200	170	ok
117-07	11	220	36	ok
117-08	12	240	92	ok
117-09	9	180	64	ok
117-10	11	220	57	ok
117-11	12	240	60	ok
117-12	11	220	45	ok
117-13	13	260	9	ok
117-14	12	240	58	ok
117-15	12	240	115	ok
117-16	10	200	65	ok
117-17	9	180	116	ok
117-18	11	220	78	ok
117-19	11	220	104	ok
117-20	12	240	111	ok
117-21	12	240	28	ok
117-22	13	260	25	ok
117-23	10	200	98	ok
117-24	11	220	130	ok
117-25	6	120	48	ok
117-26	11	220	118	ok
117-27	13	260	130	ok
117-28	12	240	98	ok
117-29	12	240	95	ok
117-30	13	260	37	ok
117-31	10	200	139	ok
117-32	10	200	63	ok
117-33	12	240	95	ok
117-34	11	220	15	ok
117-35	12	240	92	ok
117-36	11	220	87	ok
117-37	11	220	200	ok
117-38	13	260	54	ok
117-39	12	240	80	ok
117-40	12	240	183	ok
117-41	9	180	50	ok
117-42	9	180	118	ok
117-43	11	220	85	ok
117-44	9	180	69	ok